



Society of Petroleum Engineers

SPE-230027-MS

Innovative Wireline Perforation Strategy for Depleted HPHT Reservoirs with High Pressure Differentials

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Perforating in high-pressure, high-temperature (HPHT) wells require extensive engineering preparation to address hazards associated with extreme downhole conditions. Mature HPHT reservoirs often exhibit significant intra-reservoir differential pressure, in some cases exceeding 400 bar between adjacent production units. These conditions, combined with bottomhole temperatures approaching 200 °C, present high risks of gun lift, differential sticking, and tool failure, particularly in multi-zone completions where perforation order is critical. Historically, coiled tubing has been the default conveyance under live-well conditions, but in multi-zone HPHT settings it typically forces a single-run interval that can expose depleted zones to damage due to high overbalance while adding cost, complexity, and surface footprint.

This paper describes the engineering and execution of a wireline perforation programme in a deep HPHT completion with ~490-bar differential pressure across a >200-m gross interval. Conveyance was via a 0.28-in OD polymer-locked wireline cable with MP35N armour—a high-strength, sour-service design. The polymer fills between the armour wires better seals armour interstices to help block gas migration, and the small OD reduces drag and wellbore pressure forces on the cable enabling safe conveyance of 70–80 ft of 2 7/8-in guns per run.

The perforation order followed a specific pressure-profile-driven zone-by-zone sequence: pre-job simulations (gun lift, differential sticking, and shock loads) established the initial perforating plan and real-time downhole gauge pressure after each run was used to update and re-order the next sequence. Mechanical shock-loading risk was mitigated by qualifying the shock-absorber sub beyond its nominal rating. High temperature failure risk was controlled by using a flask that protected electronics at ~200 °C as well as limiting internal tool temperature build-up by rotating flask tool use. Dynamic underbalance technique was used on select intervals to improve tunnel cleanup and near-wellbore connectivity.

Across two wells, a total of 21 wireline runs (perforation plus post-perforation treatments) were successfully completed with no misruns or equipment failures and a total of ~450 m was perforated. Post-treatment pressure trends indicated improved reservoir communication. The method eliminated the need for coiled tubing, reduced operational footprint and complexity, and preserved productivity in depleted zones.

Introduction

Perforating in high-pressure, high-temperature (HPHT) wells present a unique combination of mechanical, thermal, and reservoir-related risks. In deep wells with bottomhole temperatures approaching 200°C and significant layer pressure differences, conventional perforation methods require careful adaptation. These environments raise the probability of gun lift, differential sticking, and formation damage—especially when multiple reservoir units must be accessed with varying degrees of depletion.

While coiled tubing (CT) has traditionally been used for live-well perforation in HPHT applications, it often requires a full-interval, single-run approach that limits flexibility and increases the risk of damage to depleted zones due to high overbalance. Furthermore, CT logistics increase operational complexity, footprint, and personnel requirements, which can be constrained in offshore campaigns.

This paper presents a wireline-based alternative designed for HPHT completions with large intra-reservoir differential pressure. The approach leverages advancements in cable design, simulation-based planning, and thermal mitigation to safely execute long-interval perforation runs in a staged and adaptive sequence. The method was deployed in two deep wells with differential pressure approaching 490 bar across a gross interval exceeding 200m per well. The following sections describe the engineering basis, execution workflow, and operational outcomes, demonstrating how wireline can offer a safer, more flexible, and lower-complexity alternative to CT in HPHT environments.

Conveyance Alternatives and Selection

Perforating long intervals in HPHT wells under pressure requires a conveyance method that balances mechanical strength, pressure containment, operational flexibility, and offshore deployment limitations. These include **limited deck space, crane capacity, and accommodation for additional personnel**.

Historically, **coiled tubing (CT)** has been widely used to deploy long gun strings in live-well conditions. Its ability to run and retrieve tools under pressure, while supporting circulation, makes it a robust solution in many settings. However, in deep, vertical HPHT wells with multiple reservoir zones and strong intra-reservoir pressure differences, CT presents several drawbacks:

- It typically involves a **single-pass, full-interval perforation**, reducing flexibility to optimize perforation order by pressure.
- All zones are perforated simultaneously, increasing **overbalance risk** to depleted sands and potentially causing near-wellbore damage.
- The surface footprint and equipment rig-up are **large and complex**, requiring **specialized handling, longer preparation, and higher offshore personnel-on-board (POB)**.
- Switching between logging and perforating operations involves **full rig-down and reconfiguration of pressure control equipment**, increasing non-productive time.
- Suitable weather conditions required for lifting coil onto offshore installations

In contrast, **wireline perforation** offers:

- A **smaller operational footprint**, more suitable for congested decks
- The ability to **sequence perforations zone-by-zone**, reducing overbalance exposure and improving tunnel cleanup
- Easier integration with other wireline services, such as **cement or corrosion logging**, without the need for equipment changeovers
- Lower logistical complexity, **fewer crew members**, and reduced overall POB

In this case, **wireline was selected as the optimal conveyance method** after evaluating mechanical requirements, operational risks, and rig-site limitations. The key enabler was a **0.28-in OD, polymer-filled electric line with MP35N H₂S & CO₂ corrosion-resistant armour and 6,800-lbf safe working load (SWL)**.

This solution enabled the operation to meet the well perforation objectives without the logistical and technical burdens of CT, while enhancing control over **zone-by-zone perforation exposure** and maintaining operational efficiency—without compromising safety or tool reliability. **Table 1** provides a comparative summary of the two conveyance methods evaluated for this case, highlighting the operational and logistical advantages that led to the selection of wireline over coiled tubing.

Table 1—summarizes the comparative evaluation between CT and wireline

	Deploy/retrieve guns under pressure	Optimum perforation (Zone-by-zone flexibility)	No ratihole required	Cost-effective	Surface footprint	Personnel requirement	Flexibility in sequencing	Requires full gun backup
Wireline	Yes	Yes	Yes	Yes	Minimal	Low (5 POB)	Yes	No
Coiled Tubing (CT)	Yes	No	Yes	No	Large	High (10 POB)	No	Yes

Cable Selection and Deployment Feasibility

With wireline selected as the conveyance method, the next challenge was to ensure that the cable could reliably deploy long gun strings under HPHT conditions while maintaining well control. A **0.28-in OD, polymer-filled electric line with MP35N H₂S & CO₂ corrosion-resistant armour** was selected for this campaign. The cable design provided the combination of strength, pressure sealing, and reduced drag necessary for safe perforating in a live-well environment. The cable offered several critical advantages over conventional alloy-armoured designs:

- **Strength and capacity:** The MP35N armour provided a high safe working load enabling deployment of **70–80 ft of 2 7/8-in HSD guns per run**. By comparison, conventional alloy cables were limited to ~40 ft of guns due to drag, tensile constraints, and mechanical stability risks at high pressure with PCE flow tubes. The increased run capacity directly reduced the total number of runs required, saving rig time and minimizing risk exposure.
- **Pressure sealing and gas blocking.** Conventional braided cables (left, [Figure 1](#)) contain voids between armour layers, which act as potential gas migration pathways and complicate sealing at the wellhead. Grease injection is required to block these pathways, but sealing efficiency is more difficult at high wellhead pressures, particularly in sour service gas wells, where there is a higher risk of cable damage under high pressure.

The polymer-filled MP35N cable (right, [Figure 1](#)) eliminated these pathways with a **void-free structure** in which polymer fills all interstitial spaces between conductor and armour. This design helps to block internal gas migration, improves sealing integrity at the stuffing box, reduces grease requirements, and enhances safety in HPHT live-well conditions.

- **Reduced drag and improved stability:** The small 0.28-in OD reduced cable drag both in the wellbore and across flow tubes, as hydraulic forces increase with larger cross-sectional area. The torque-locked construction minimized rotation, bird-caging, and stretch, eliminating the need for

seasoning runs and reducing the risk of armour instability (armour torque imbalance), especially under high pressure at surface with PCE. Importantly, the **friction coefficient observed during the job was at ~ 0.166** , significantly lower than values observed in previous campaigns using a larger 0.32-in OD cable.

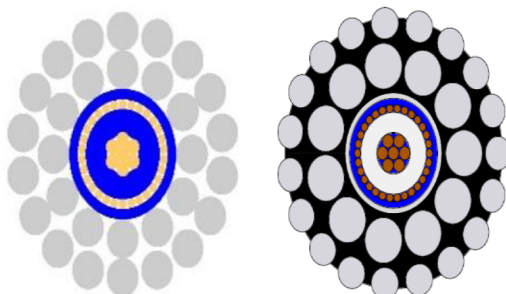


Figure 1—Comparison of cable structures. Left: conventional cable with unfilled voids that create potential gas migration pathways. Right: polymer-filled MP35N cable with void-free structure sealing armour interstices and blocking gas migration.

Validation of deployment feasibility

Pre-job tension simulations were conducted for each planned run, confirming that toolstring could be deployed and retrieved within cable safe operating limits. Post-job data demonstrated close agreement between measured and simulated surface tension, with an **overpull margin of $\sim 2,000$ lbf at total depth**. The **friction coefficient of ~ 0.166** observed during the job, significantly lower than values observed in previous campaigns with 0.32-in OD cable. This validation confirmed the accuracy of the modelling assumptions and the suitability of the cable for HPHT perforating operations (Figure 2).

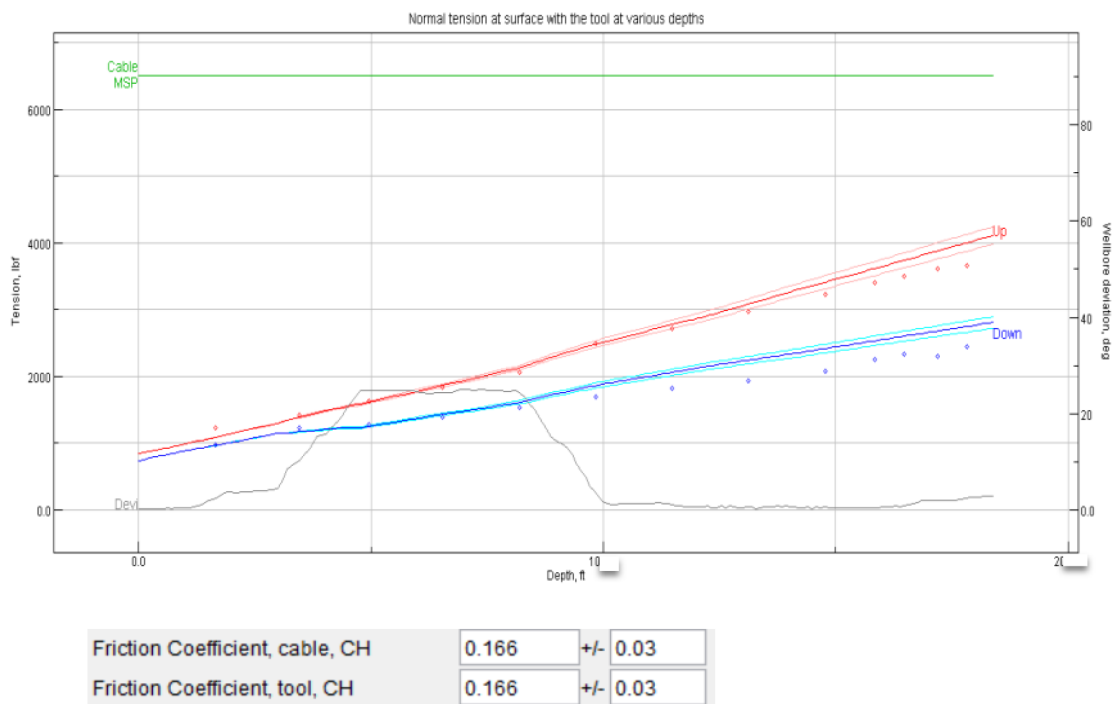


Figure 2—Comparison of simulated vs. measured surface tension. The actual surface tension (dots) closely matched simulated curves (solid lines) during deployment and retrieval and remained below the maximum safe pull (MSP), validating pre-job modelling and confirming safe operational margins.

Following execution, the measured surface tension profiles closely matched the simulated curves, confirming the validity of the planning assumptions and the cable's performance in the well. This alignment

between model and field data provided assurance that the cable could support long gun deployments safely and efficiently under extreme HPHT conditions.

Release Devices and Contingency Planning

In wireline perforation, a weakpoint provides the controlled release mechanism in the event of a stuck gunstring, preventing cable parting at surface and leaving a fishing neck accessible for recovery. The choice of weakpoint was critical for this HPHT campaign, given the depth and heavy toolstring weight.

- **Mechanical weakpoint:** For this application, the maximum usable rating was approximately 1,800–2,150 lbf to ensure it could break at total depth with a 70–80 ft gunstring. This limited capacity did not provide a sufficient overpull margin for safe recovery in case of sticking and was therefore considered unsuitable.
- **Electrical weakpoint:** An electrical weakpoint was selected instead. The device had a tensile strength rating of 12,000 lbf and a maximum operating temperature of 260 °C. This provided a high overpull margin at total depth while maintaining the ability to be activated by electrical command from surface if the toolstring became stuck.

As a secondary contingency, a **cable cutter** was mobilized. In scenarios where communication with the weakpoint could not be established (e.g., cable damage), the cutter could be gravity-deployed along the line to sever the cable above the toolstring. The cutter attaches to the severed cable end, allowing retrieval to surface and leaving behind a clean fishing neck for subsequent recovery. Programmable timers were also available to ensure activation in case of failed impact triggering.

Together, the electrical weakpoint and cable cutter provided a robust contingency plan to manage stuck-tool risk during live-well HPHT perforation.

Thermal Management Strategy

Bottomhole static temperatures reaching nearly 200 °C, required the use of flasks to protect sensitive electronic components. Two critical tools were flaked: the **electronics board for the weakpoint release system** and the **gamma ray (GR) correlation tool**. Both would have been highly vulnerable to rapid degradation or failure if directly exposed to these conditions.

On surface, two approaches are available for cooling flaked tools:

1. **Immediate opening of the flask** after retrieval to accelerate cooling. However, if the tool remains hot, this exposes the electronics to uncontrolled **thermal shock**, risking component failure.
2. **Cooling in the flask**, which provides slower but controlled temperature reduction and avoids shock loading of electronics.

For this operation—planned for ~14 runs (includes a spent gun run)—the second option was selected. This method, known as the **staged cooling approach**, was implemented as follows:

- Three flaked tools were mobilized offshore.
- The flaked GR tool and weakpoint electronics were rotated between runs (Flask #1 for Run 1, Flask #2 for Run 2, Flask #3 for Run 3, then back to Flask #1 for Run 4, etc.).
- Each flaked tool was allowed **>24 hours of controlled cooling** before re-use.
- Cooling was performed without removing the electronics from the flask, preventing uncontrolled exposure.

This approach **minimized thermal shock**, protected both the weakpoint electronics and GR correlation system, and crucially **increased operational efficiency**. By rotating flaked tools, there was no need to wait for cooling or replace electronics between runs—another set of tools was always ready at surface. At

the same time, the staged cooling workflow **maintained tool reliability over multiple HPHT exposures**, ensuring the weakpoint release system and GR correlation tool remained functional throughout the 14-run programme.

Shock Absorber Deployment

Detonation of long gun strings generates high transient axial loads that can damage cable heads and sensitive electronics if unmanaged. In this campaign, each 70–80 ft gun string weighed ~971 lb, A **load test to 1,000 lbf** was performed prior to the job, confirming that the absorber could sustain the static gun weight while still providing effective damping of detonation shock. During operations, shock absorbers were **inspected and redressed after each run**, ensuring consistent performance across all runs.

The use of a shock absorber was critical for two reasons:

- **Tool protection:** It reduced detonation shock loading on the cable head and protected the flaked electronics (weakpoint release board and GR tool), both of which were highly sensitive to mechanical stress.
- **Operational reliability and efficiency:** By controlling shock loads, the absorber minimized the risk of premature tool failure, allowing the program to proceed through 14 perforation runs without interruption. This helped maintain tool integrity and avoided delays associated with equipment replacement or troubleshooting.

The deployment of a shock absorber was therefore a key enabler in sustaining both **equipment reliability** and **operational efficiency** under extreme HPHT perforating conditions.

Perforation Planning and Sequencing

The principal reservoir challenge was the **large differential pressures between the reservoir layers**, approaching up to **490 bar** between adjacent units. A conventional bottom-up perforation program would have exposed depleted zones to excessive overbalance, with high risk of **gun lift, differential sticking, and near-wellbore damage**. To manage these risks, the operation was designed as a **zone-by-zone perforation sequence**, guided by both pre-job simulations and real-time pressure monitoring.

The reservoir was divided into three main flow units (Units 1–3), each exhibiting distinct pore pressures as shown in [Table 2](#) and [Figure 3](#). Over time, Units 2 and portions of Unit 1 became significantly depleted, while other intervals remained at much higher pressures. The result was a strong **intra-reservoir differential pressure of up to 490 bar**.

Table 2—Reservoir Units and Pore Pressure

Depth	Zone	Reservoir Unit	Pore Pressure (Bar)
Shallowest	1	Unit 3	480
increase in depth	2	Unit 2 Upper	350
	3	Unit 2 Lower	185
	4	Unit 1 Upper (a)	460
	5	Unit 1 Upper (b) (faulted)	675
Deepest	6	Unit 1 Lower (a)	240

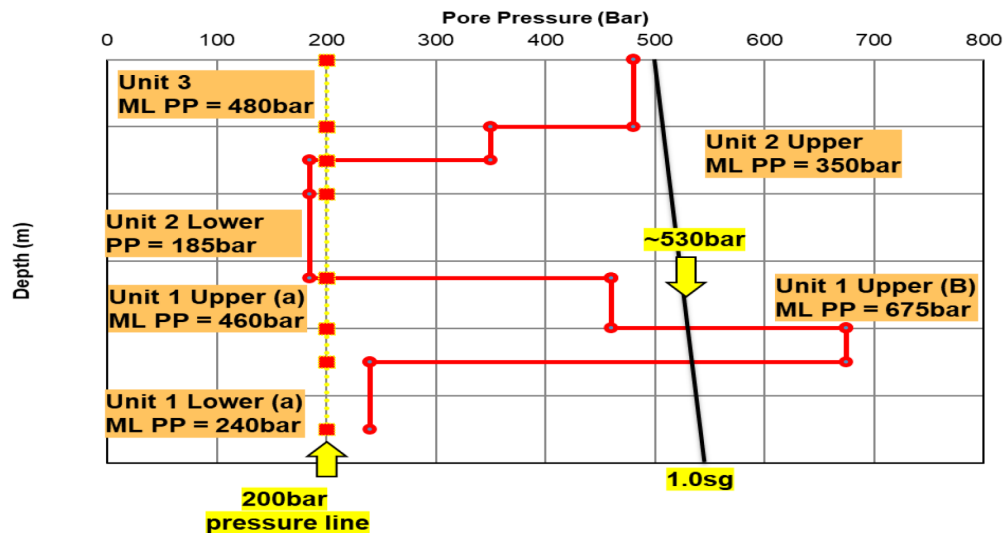


Figure 3—Pore pressure profile by depth.

The profile shows strongly depleted intervals (Unit 2 Lower ~185 bar, Unit 1 Lower ~240 bar) adjacent to higher-pressure compartments (Unit 3 ~480 bar, Unit 1 Upper (a) ~460 bar) and a fault-sealed high-pressure sub-compartment (Unit 1 Upper (b) ~675 bar). The wellbore hydrostatic line (~530 bar) highlights (i) extreme overbalance in the most depleted units and (ii) significant underbalance in the faulted high-pressure unit, explaining the twin hazards of differential sticking / damage and gun lift if a standard bottom-to-top sequence were used.

Conventional Bottom-to-Top Perforation Sequencing: Risk Assessment

A conventional approach would be to perforate **from bottom to top**. This traditional bottom-to-top perforation sequence was initially considered and fully simulated. However, it was **not selected** due to multiple operational risks identified during modelling. The well had a complex pressure profile [Figure 4](#), with some **heavily depleted zones** and others still at **high pressure**, including one fault-isolated compartment. A fixed-density fluid (~1.05 sg seawater) gave an initial **bottomhole pressure (BHP)** of ~530 bar.

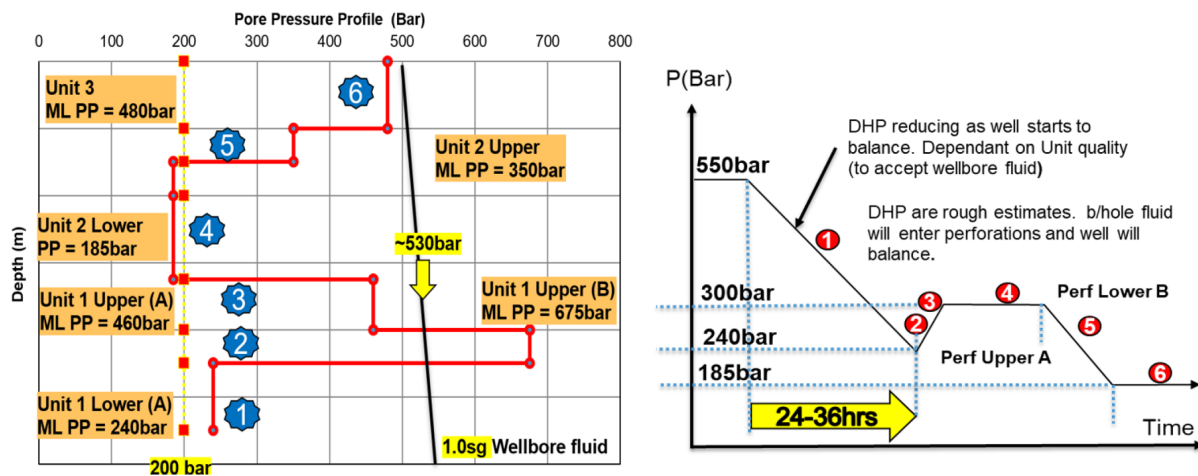


Figure 4—Most likely pore pressure (ML PP) by reservoir unit (left) and predicted downhole pressure trajectory during the initial bottom-to-top perforation sequence (right).

Using the conventional bottom-to-top perforation sequence, the first shot would be **Unit 1 Lower (A)** at **240 bar** against a **~550 bar BHP**—about **+310 bar overbalance**—creating prime conditions for **differential sticking** and **near-wellbore damage**. The next step would immediately swing to the opposite extreme: the fault-isolated **Unit 1 Upper (B)** at **675 bar**, or **–435 bar underbalance**, with a high likelihood of **gun lift/tool movement**. The remaining targets still sit far from the wellbore pressure, so the sequence alternates between **large overbalance and large underbalance**. For this reason, the conventional bottom-to-top approach **was operationally high risk**.

These risks are summarized in [Table 3](#) and the predicted evolution of wellbore pressure is shown in [Figure 4](#) (left). The modeling demonstrated that a simple bottom-to-top sequence would expose depleted units to damaging overbalance and high-pressure units to extreme underbalance, making it unsuitable for safe or efficient execution in this reservoir.

Table 3—Risk assessment for a conventional bottom-to-top sequence

Perf order	Zone / Unit	ML PP (bar)	Assumed BHP before shot (bar)	ΔP (+OB / – UB) (bar)	Risk summary
1	Unit 1 Lower (A)	240	550	+310 (OB)	High overbalance: likely differential-sticking / tunnel damage
2	Unit 1 Upper (B) (faulted, high-P)	675	240	–435 (UB)	High underbalance: Gun-lift/tool-movement risk
3	Unit 1 Upper (A)	460	240	–220 (UB)	Underbalance: (high-perm streak), Gun-lift/tool-movement risk .
4	Unit 2 Lower	185	~300	+115 (OB)	High overbalance; likely differential-sticking / tunnel damage .
5	Unit 2 Upper	350	185	–165 (UB)	Underbalance: Lower risk of gun lift at this ΔP due to low permeability, however, high perm streak cannot be discounted and thus risk of gun lift .
6	Unit 3	480	185	–295 (UB)	High Underbalance : high -perm intervals cannot be shot safely until wellbore pressure rises (Gun-lift/tool-movement risk .)

Optimized Perforation Sequencing Plan

To mitigate the risks identified in the conventional bottom-to-top perforation sequencing, an **adaptive sequence** ([Figure 5](#)) was implemented with primary objectives: (i) use the faulted high-pressure unit to **pressure-up the system** early, and (ii) approach depleted intervals only after the wellbore hydrostatic had moved to safer differentials. The decision process combined pre-job simulations with **live downhole pressure gauges** monitoring after every run.

The sequencing objective was to keep the **pressure differential (ΔP)** for each perforation interval within a **narrow operating window** to reduce identified risks. In practice this meant: **(i)** avoiding large **overbalance (OB)** on the most depleted units to limit sticking and near-wellbore damage, **(ii)** limiting **underbalance (UB)** in the faulted high-pressure compartment to prevent gun lift, and **(iii)** using each perforation step to **improve conditions** for the next perforation run. The plan was built from a **Most Likely Pore Pressure (ML PP)** profile from a robust reservoir model based on offset well RFT data and a **wellbore-pressure forecast that predicted the BHP changes after each perforation sequence**. The BHP forecast would be compared to actual BHP from DHG data during the perforating operation. Initial BHP before any perforation was **~530 bar** (≈ 1.0 sg inhibited brine).

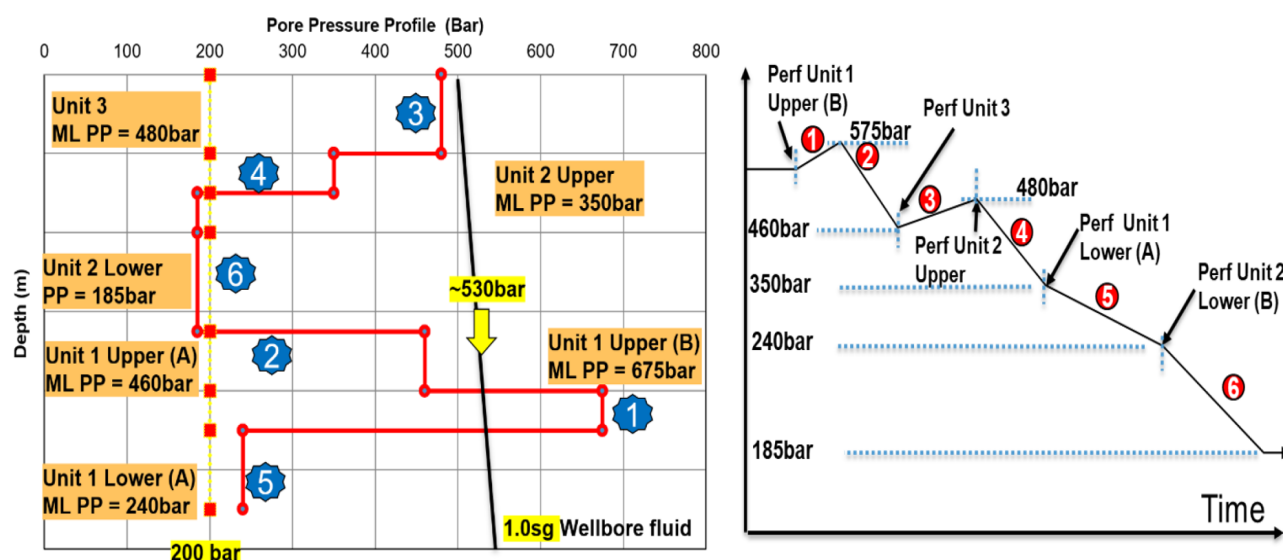


Figure 5—Optimized sequence: (Left) ML PP profile vs the perforation order; (Right) predicted BHP trajectory vs perforation sequence vs time.

Sequence Design Principles

- **Open the faulted high-pressure unit early** to raise system BHP and reduce OB exposure in later depleted shots.
- **Target moderate-pressure units next** (e.g., Unit 1 Upper (A), Unit 3) to establish communication and validate the forecast without extreme ΔP .
- **Defer depleted sands** (Unit 1 Lower (A), Unit 2 Lower) **until the wellbore pressure has stabilized within the planned overbalance window for those intervals**; then shoot at the **lowest practical OB**
- After each run, Downhole pressure **gauge-measured BHP** was compared to the forecast if outside, the **next target was re-ranked** (e.g. swap steps, shorten gun length) to keep ΔP within limits.

Table 4—optimized, gauge-driven perforation sequence (order, ML PP, modelled BHP pre-shot, ΔP , and rationale).

Order	Zone / Unit	ML PP (bar)	Assumed BHP pre-shot (bar)	ΔP (+OB/-UB)	Rationale
1	Unit 1 Upper (B), faulted (HP)	675	530	-145	Open HP unit first to raise system BHP ; short guns; close surveillance to limit movement.
2	Unit 1 Upper (A)	460	575	115	Moderate OB; extend connectivity within Unit 1 under stable conditions.
3	Unit 3	480	460	-20	Small UB; rapid balance with Unit 1; cross-check model vs gauge.
4	Unit 2 Upper	350	480	130	Acceptable OB once BHP is elevated by steps 1–3.
5	Unit 1 Lower (A)	240	350	110	First depleted sand: Post perforation treatment with dynamic underbalance planned post-shot.
6	Unit 2 Lower	185	240	55	Most depleted; shot last at lowest practical OB; Post perforation treatment with dynamic underbalance planned post-shot.

Treatment of the most depleted units

Treatment of the Most Depleted Units. Unit 1 Lower (a) and Unit 2 Lower, the most depleted intervals, were intentionally deferred and perforated only after the pressure differential (ΔP) to these sands was minimized, targeting a reduced overbalance window of approximately +60 to +110 bar. Immediately post perforation treatment, each interval received a Dynamic Underbalance (DUB) pressure pulse to remove debris and fines, aiming to restore near-wellbore connectivity. Gauge trends post-DUB shifted from declining to stable or increasing bottomhole pressure (BHP), aligning with the predicted pressure profile and indicating that the DUB pulse effectively cleaned the perforation tunnels.

Wellbore Dynamics and Shock Simulation

To assess the pressure transient response of the gunstring and evaluate potential risks associated with high-pressure differential conditions, a full dynamic perforation simulation was performed using Simulation software. The software uses first-principles physics to solve the governing equations for conservation of mass, momentum, and energy, accounting for coupled effects of **wellbore pressure waves, reservoir inflow, shock absorber behavior, and gunstring/cable dynamics** (Baumann et al. 2011, 2015).

The objective of this simulation was threefold:

1. To evaluate **axial shock loads** on the cablehead and other weak points
2. To ensure that gun movements are acceptable and no guns are blown uphole
3. To assess **wellbore pressure dynamics**, particularly to predict the stabilized borehole pressure after each perforation.

Simulation Response — Unit 1 Upper (B)

The simulated event represents **Run 1**, where **Unit 1 Upper (B)** (estimated pore pressure: 675 bar) was perforated under a **145 bar dynamic underbalance** condition. The wellbore was filled with **1.0 sg fresh water**, resulting in a pre-shot hydrostatic pressure of **~530 bar** perforation depth.

Three primary outputs are shown in [Figure 6](#):

- **Plot A — Transient wellbore pressure:** Following detonation, the wellbore pressure dropped below reservoir pressure due to DUB, then stabilized at ~530 bar after ~1.1 seconds, consistent with hydrostatic expectations for this fluid density.
- **Plot B — Cablehead force:** The initial load was ~0.4 tons (~900 lbf), which decreased to ~0.2 tons (~450 lbf) immediately post-shot, before peaking at ~0.75 tons (~1650 lbf) due to downward acceleration. The peak shock load remained within ~60% of the weak point's safe working load, indicating no risk of failure.
- **Plot C — Gun string axial movement:** The gun string experienced an initial upward displacement of ~10 cm within 0.1 sec of detonation, followed by a progressive downward movement of ~65 cm at 1.2 sec. This displacement remained within acceptable mechanical limits for the deployed configuration.

The simulation confirmed that **shock-induced loads and gun string movements remained within safe design limits** and predicted **no mechanical integrity issues** when perforating Unit 1 Upper (B) under –145 bar underbalance. Equivalent simulations were performed for all target reservoir units, ensuring that perforation could be executed safely across the entire campaign.

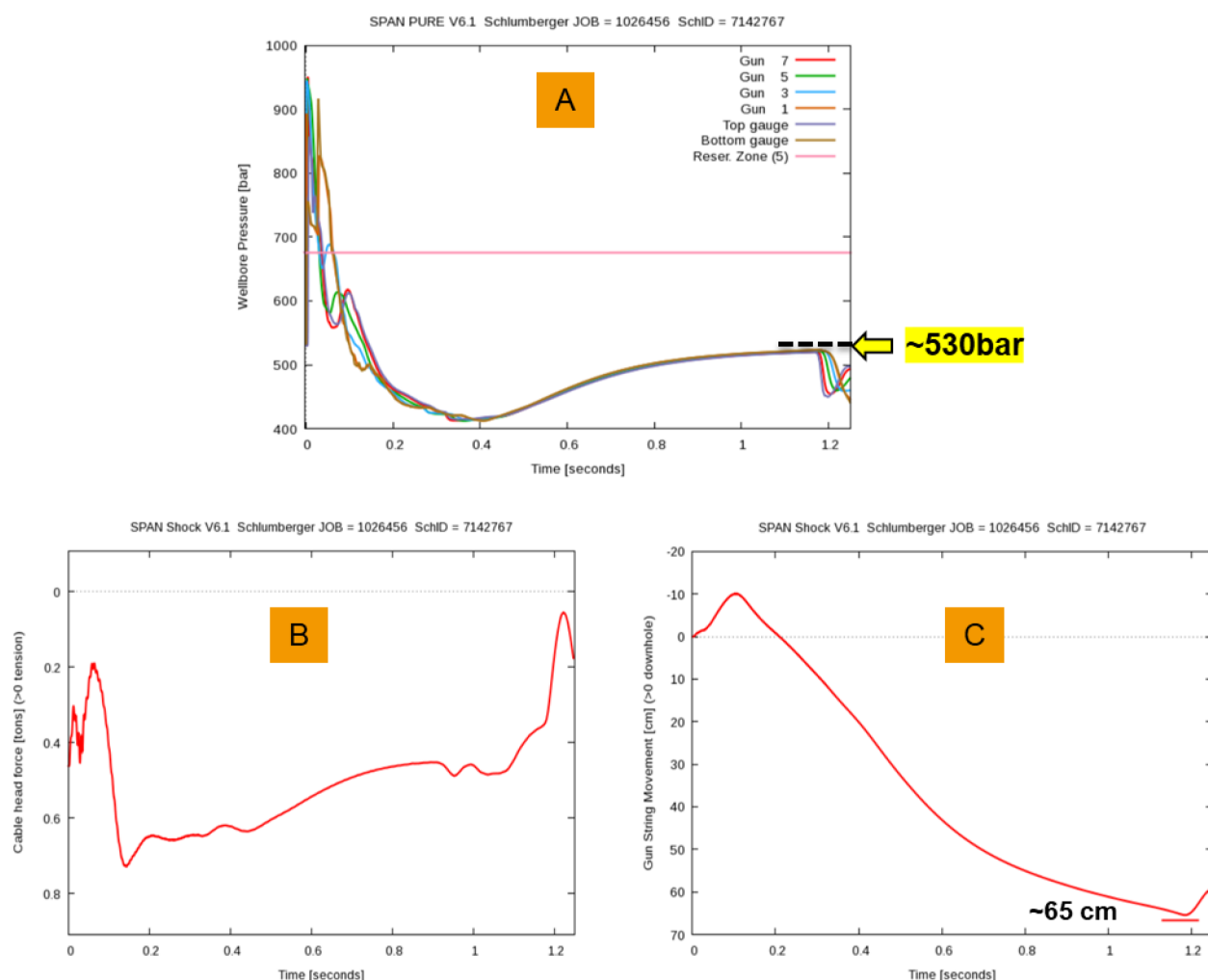


Figure 6—Wellbore pressure, cablehead force, and gunstring movement from simulation (Unit 1 Upper B, ~145 bar DUB).

Field Execution and Results

The perforation campaign was executed in a complex HPHT well environment characterized by sour gas, multiple reservoir units with varying pressure regimes, and significant drawdown in several zones. A total of 14 runs were completed, comprising 10 perforation runs, 3 post-perforation Dynamic Underbalance (DUB) treatment runs.

Perforation was performed using 2-7/8 in HNS guns, all 70 ft in length, across six distinct reservoir intervals. Each unit exhibited unique pressure behavior, with Most Likely Pore Pressure (ML PP) estimates ranging from 185 bar to 675 bar. A **borehole fluid weight of 1.0 sg (water)** was maintained in the wellbore **prior to perforation**, resulting in a **hydrostatic pressure of approximately 530 bar** at reservoir depth. The gun sequence and depth strategy were aligned with pre-job simulations and continuously optimized during execution to minimize the risk of tool lift or gun blowout—particularly when transitioning between high- and low-pressure zones. Real-time adjustments to perforation sequencing and overbalance modeling were made in response to BHP trends and wellbore response after each perforation run.

The downhole pressure trend, recorded via a permanent Downhole gauge is shown in Figure 7. Perforation runs (green markers 1–10) were followed by a period of continued pressure decline, particularly after Run 10. This behavior suggested restricted connectivity or tunnel debris accumulation, especially across the two lowest-pressure zones: Unit 1 Lower (A) and Unit 2 Lower.

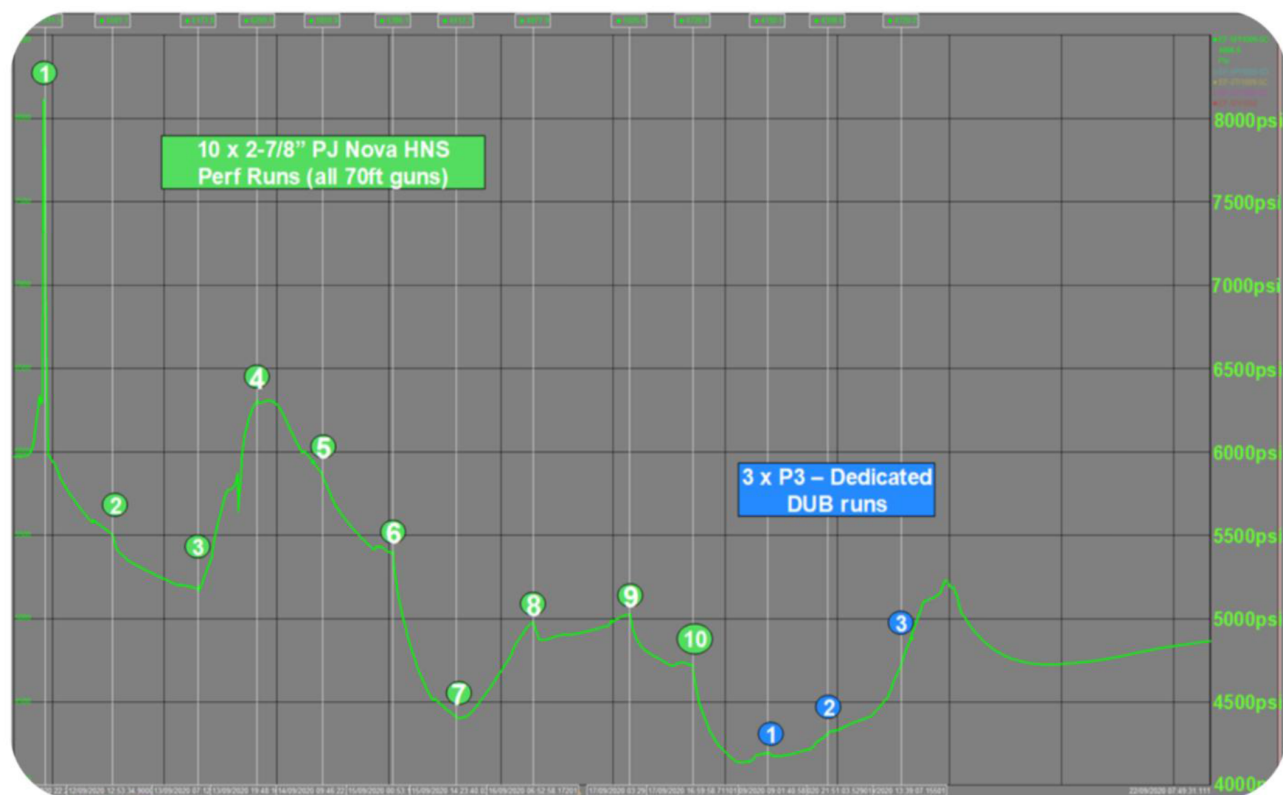


Figure 7—Bottomhole pressure response from the permanent downhole pressure gauge, showing the effect of perforation runs (green markers 1–10) and post-perforation DUB treatments (blue markers 1–3). The increasing pressure trend after DUB confirms restored near-w

To restore wellbore communication and reduced the effect on overbalance perforation on the most depleted unit, three dedicated post-perforation DUB treatments runs (blue markers 1–3 in Figure 7) were executed. Each run applied a dynamic underbalance pulse to clean perforation tunnels and improve near-wellbore flow. Following these treatments, the bottomhole pressure trend shifted from declining to gradually increasing—a clear indication of successful tunnel cleanup and improved inflow from the perforated zones.

This post-treatment pressure recovery is in excellent agreement with pre-job simulation forecasts and validating the effectiveness of the perforation and DUB strategy.

Discussion

This campaign demonstrated that a wireline-based approach can be safely and effectively applied in HPHT environments with significant intra-reservoir pressure differentials. The use of a 0.28-in polymer-filled MP35N cable was a key enabler, offering the mechanical capacity and low friction necessary to deploy 70–80 ft gun strings with reliable pressure control. Its void-free, gas-blocking construction provided additional assurance against gas migration—an essential consideration in sour wells with very high wellhead pressures. Field measurements confirmed that cable tension closely matched the pre-job model, validating the assumed friction coefficient (≈ 0.166) and preserving an ample overpull margin throughout operations.

Perforation sequencing was not treated as a static design but as a dynamic process driven by real-time data. Each run was evaluated against measured bottomhole pressure (BHP), and the plan was updated accordingly to keep ΔP within a narrow operating window. This adaptive approach proved especially valuable in managing the transition between high-pressure and depleted zones, avoiding extreme overbalance or underbalance conditions that could trigger gun lift, tunnel damage, or differential sticking.

Thermal management was another critical factor. With bottomhole temperatures approaching 200 °C, flanked housings and staged tool rotation enabled reliable gamma-ray correlation and electronic weakpoint

operation without thermal degradation. Likewise, shock mitigation through a wireline absorber successfully dampened detonation loads and prevented shock-related anomalies, even with an 80-ft gun payload.

Where overbalance into depleted intervals could not be entirely avoided, dynamic underbalance (DUB) was applied as a targeted post-treatment. The measured BHP response—marked by a transition from decline to a stable post-DUB plateau—confirms that this treatment enhanced near-wellbore connectivity. The effectiveness of DUB in restoring productivity in the lowest-pressure zones highlights its value as a complement to ΔP -managed sequencing.

Collectively, these results support the broader applicability of this workflow—polymer-filled wireline, adaptive BHP-driven sequencing, thermal and shock mitigation, and post-shot DUB—in other HPHT completions with complex reservoir architecture.

Conclusions

This work demonstrates that a carefully engineered wireline perforation strategy can safely and effectively address the challenges of long-interval HPHT completions with large intra-reservoir pressure differentials. Central to this approach was the use of a pressure-sequenced, gauge-driven workflow that adapted the perforation order based on real-time bottomhole pressure (BHP) measurements. By sequencing from high-pressure to low-pressure units—opening the fault-isolated compartment early and deferring the most depleted zones—extreme overbalance and underbalance conditions were avoided, thereby mitigating risks such as gun lift, differential sticking, and near-wellbore damage.

Real-time pressure feedback after each run enabled continuous optimization of the firing sequence, ensuring that the pressure differential (ΔP) remained within a safe, defensible window throughout the operation. Post-shot dynamic underbalance (DUB) treatments further improved tunnel cleanup and enhanced near-wellbore connectivity, particularly in the most depleted sands where drawdown could not be avoided.

Thermal management was critical to operational continuity. With bottomhole temperatures near 200 °C, the use of flanked correlation and release tools—combined with a staged rotation protocol—ensured electronic integrity without compromising efficiency. Likewise, shock mitigation techniques successfully isolated the cablehead from detonation loads, even with extended gun payloads.

Across two HPHT wells, the program completed 21 runs, perforating approximately 450 meters without misruns or HSE incidents. The observed pressure response—particularly the transition to a stable post-treatment plateau following DUB—confirmed that reservoir communication was preserved and enhanced. This method provides a transferable template for high-complexity HPHT perforation, offering zone-by-zone control, reduced operational footprint, and improved access to depleted intervals without reliance on coiled tubing.

Nomenclature

BHP	= bottomhole pressure
CT	= coiled tubing
DUB	= dynamic underbalance
GR	= gamma ray
HPHT	= high-pressure, high-temperature
ML PP	= Most-Likely Pore Pressure
MSP	= maximum safe pull
OB	= overbalance
PCE	= pressure-control equipment
POB	= personnel on board
SWL	= safe working load

UB = underbalance

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